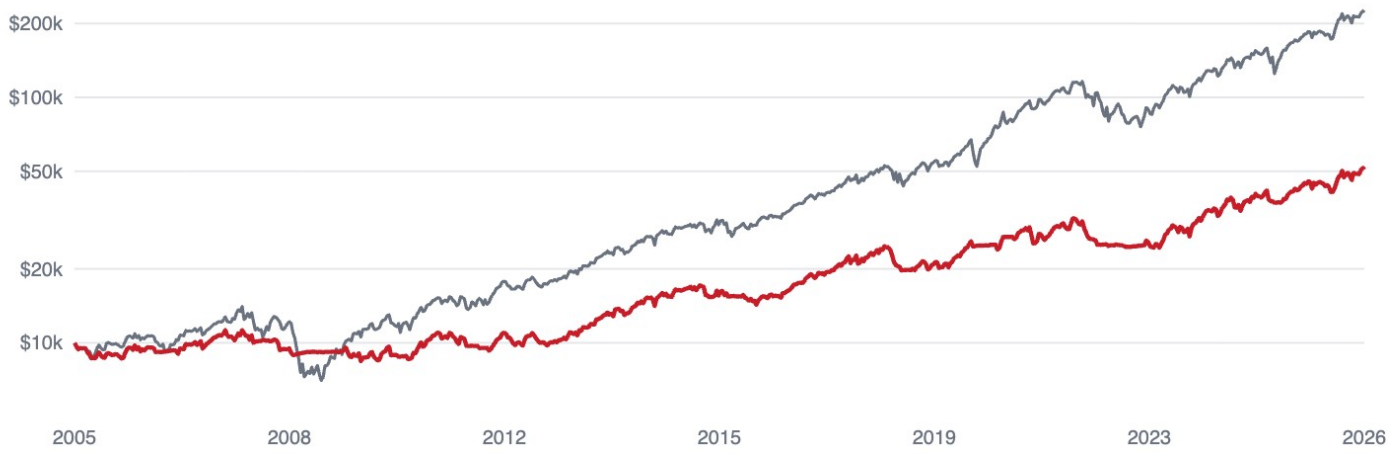
QQQ trend filter with VIX guard ★★☆☆☆ 2026-10-14

CAGR	Total Return	Sharpe	Max Drawdown	Volatility	Quant Edge	Uncertainty	Time Invested
+7.8%	+407%	0.62	-26.3%	13.6%	None	Medium	74%
Bench +15.3%	Since 2005-01	Bench 0.77	Bench -53.4%	Annualised	3 of 22 yrs ahead	1-yr range 44%	116 trades

Growth of $10,000 vs QQQ (log scale)



2020	2021	2022	2023	2024	2025	2026	Year
+22.7%	+7.5%	-19.8%	+30.6%	+21.9%	+12.1%	+14.5%	Strategy %
+48.4%	+27.4%	-32.6%	+54.9%	+25.6%	+20.8%	+21.5%	QQQ %

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About this report

Rules were written by Marvell from your description and tested on daily prices. The Quant Rating compares risk-adjusted return (Sharpe) with the benchmark; Edge counts the calendar years it came out ahead; Uncertainty is the width of the one-year simulated range.

Two-Star QQQ Trend Filter With VIX Guard Trails Benchmark

Quant Note Marvell, MercerFargo Quant, 2026-10-14

From 2005 to 2026, the strategy returned 7.8% CAGR versus 15.3% for QQQ, with lower volatility (13.6% vs 21.6%) and a smaller max drawdown (-26.3% vs -53.4%). It beat the benchmark in 3 of 22 years, and the derived gap is -7.6 CAGR points and -1707 total-return points. Rating 2/5, edge None, uncertainty Medium.

Why it matters:

The profile is defensive: max drawdown was 49% of the benchmark's and volatility 63% of the benchmark's, but the return shortfall was material. For someone considering this approach, the VIX/200-day guard may cut drawdown without keeping pace with QQQ.

› Max drawdown -26.3% vs -53.4% for QQQ.

› Volatility 13.6%, or 63% of the benchmark's.

The bottom line:

A lower-risk but return-lagging overlay in this backtest; past results and simulations are not predictions.

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Strategy Description

Hold QQQ while it trades above its 200-day moving average and the VIX is below 25; otherwise park the capital in short-term Treasuries (SHY).

- Long only, fully invested in QQQ when in a trade, otherwise all capital sits in SHY
- VIX is read from the ^VIX daily close and compared with the same close as QQQ
- Both entry conditions must be true to enter and either failing exits at the next close
- Signals act at the close and returns accrue from the following day
- QQQ total-return adjusted closes are used for the trend test
- Not modelled: Intraday VIX spikes are not modelled; only end-of-day VIX is used. SHY cash yield is captured through its price return, not as an explicit interest rate.
- Tested 2005-01-03 to 2026-10-13 with 5 bps cost per side; benchmark QQQ bought and held.
- Parameters: Trend moving average (days) 200, Max VIX to hold QQQ 25.

Strengths

- Max drawdown -26.3% vs -53.4% for QQQ.
- Volatility 13.6%, or 63% of the benchmark's.
- Beta 0.38, alpha 2.0%/yr, Sortino 0.85, and profit factor 2.21.

Risks

- CAGR lagged by 7.6 points and total return by 1707 points versus QQQ.
- Win rate 30% across 116 trades; beat the benchmark in only 3 of 22 years (14% of years), with edge None and rating 2/5.
- Not modelled: intraday VIX spikes, and SHY cash yield is captured via price return rather than explicit interest; only end-of-day VIX is used, so live results could differ.

Projection Drivers

The projection from $10,000 over 5 years spans $8.9k at the 5th percentile, $14k median, and $24k at the 95th, with a 10% chance of losing money. With return shift 0%/yr and volatility x1, the median is 44% total gain (7.5%/yr), 5th percentile -11% (-2.3%/yr), and 95th 135% (18.6%/yr); changing return-shift or volatility assumptions moves those outcomes. The projection resamples the strategy's own daily returns in 21-day blocks over 5 years (400 paths).

Risk and Uncertainty

Uncertainty is rated Medium: in the simulation, one year from now the 5th to 95th percentile range spans about 44% of starting capital. The deepest historical fall was -26.3% (benchmark -53.4%), and the longest stretch below a previous high lasted 1505 trading days. Beta to QQQ is 0.38 with correlation 0.61.

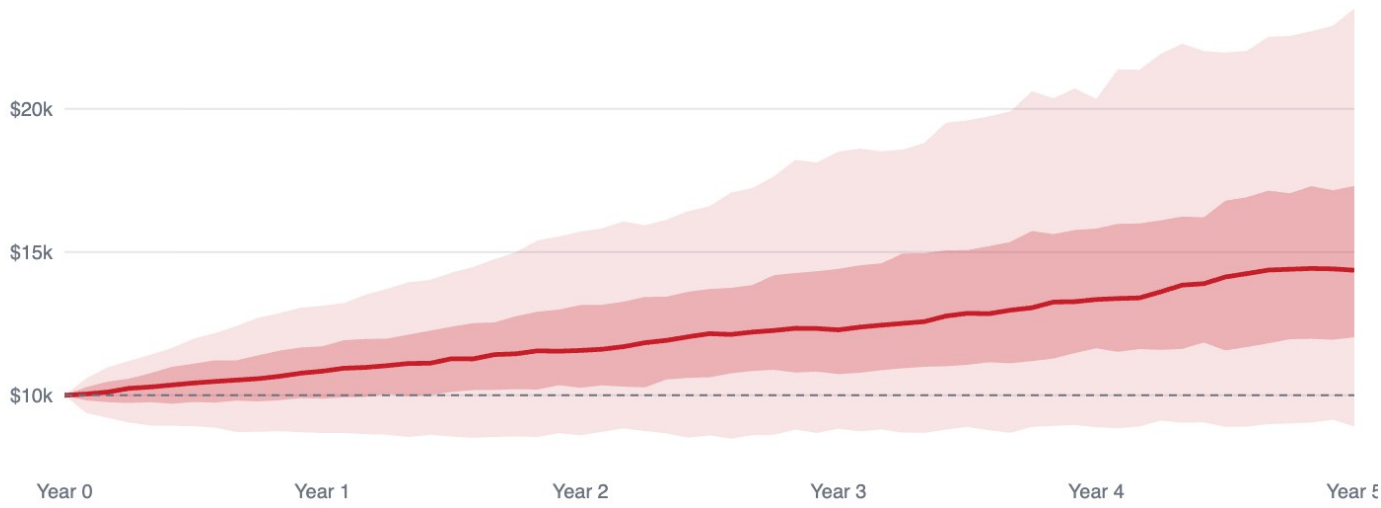
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Projection: value of $10,000 over 5 years



Percentile	5th	25th	Median	75th	95th
Ending value	$8.9k	$12k	$14k	$17k	$24k
Return per year	-2.3%	+3.8%	+7.5%	+11.6%	+18.6%

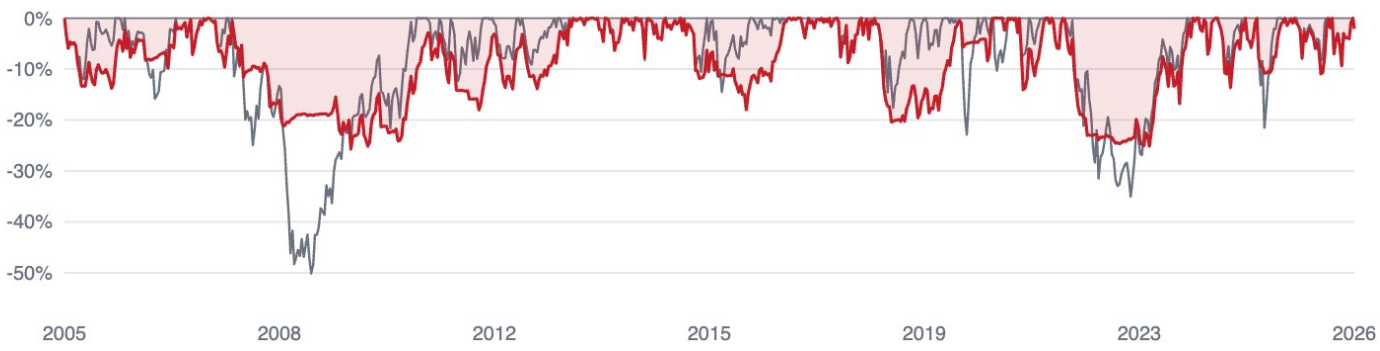
Chance of ending below $10,000: 10%

Risk Statistics

	Strategy	QQQ
CAGR	+7.8%	+15.3%
Total return	+407%	+2114%
Volatility	13.6%	21.6%
Sharpe ratio	0.62	0.77
Sortino ratio	0.85	1.10
Max drawdown	-26.3%	-53.4%
Calmar ratio	0.30	0.29
Best year	+36.6%	+54.9%
Worst year	-19.8%	-41.7%

	Strategy
Beta	0.38
Alpha (per year)	+2.0%
Correlation	0.61
Time invested	74%
Trades	116
Win rate	30%
Average trade	+1.7%
Average holding (days)	34
Years ahead of benchmark	3 of 22

Drawdown (% below previous high)



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Calendar-Year Returns (strategy red, QQQ grey)



Monthly Returns (%)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2026	1.2	-2.3	-5.8	10.2	10.6	-0.1	-6.6	4.2	3.3	0.5		
2025	2.2	-2.7	-5.6	0.8	1.5	6.4	2.4	1.0	5.4	3.9	-2.4	-0.7
2024	1.8	5.3	1.3	-4.4	6.2	6.5	-1.7	-2.2	2.6	-0.9	5.4	0.8
2023	-0.5	-0.4	2.6	0.5	7.9	6.3	3.9	-1.5	-5.1	-2.1	10.8	5.6
2022	-9.3	-5.7	-2.8	-2.7	0.6	-0.6	0.4	-0.8	-1.2	-0.1	0.7	0.1
2021	-0.3	-4.2	-5.3	5.9	-2.1	6.3	2.9	4.2	-6.0	7.9	-0.4	-0.2
2020	3.0	1.6	1.2	0.3	-0.1	0.6	-2.8	10.8	-0.0	-0.0	2.1	4.4
2019	0.3	0.5	1.8	5.5	-8.2	4.4	2.3	-1.9	0.9	4.4	4.1	3.9
2018	8.8	-2.6	-4.1	0.5	5.7	1.1	2.8	5.8	-0.3	-13.7	-3.8	-3.4
2017	5.1	4.4	2.0	2.7	3.9	-2.3	4.1	2.1	-0.3	4.6	2.0	0.6
2016	-2.8	0.1	0.0	-2.7	0.3	-6.2	7.2	1.1	2.2	-1.5	0.4	1.1
2015	-2.1	7.2	-2.4	1.9	2.2	-2.5	4.6	-8.7	-1.6	4.6	0.6	-1.6

Trade Log (latest 10 of 116)

Entry	Exit	Return	Days	Exit reason
2026-04-08	open	+22.9%	130	open
2026-03-19	2026-03-20	-1.8%	1	signal
2026-03-16	2026-03-18	-0.9%	2	signal
2026-03-10	2026-03-12	-1.7%	2	signal
2025-11-21	2026-03-06	+1.8%	70	signal
2025-10-17	2025-11-20	-3.0%	24	signal
2025-05-12	2025-10-16	+18.4%	109	signal
2025-03-25	2025-03-26	-1.8%	1	signal
2024-12-19	2025-03-06	-4.9%	50	signal
2024-08-08	2024-12-18	+15.4%	92	signal

Appendix: Research Methodology

Data: daily closes adjusted for splits and dividends. Execution: a rule is checked at each close and the position changes at that close, earning from the next day, so no future information is used. Costs: the stated basis points on every unit traded, each way; idle cash earns nothing. Metrics: CAGR and volatility are annualised from daily returns (252 days); Sharpe and Sortino use a 0% risk-free rate; drawdowns are measured from the running high. Quant Rating: 5 stars when the Sharpe ratio beats the benchmark's by more than 0.4, 4 by 0.15, 3 within ±0.15, 2 behind by up to 0.4, 1 below that. Edge: Wide when ahead of the benchmark in at least 70% of calendar years, Narrow at 55%, otherwise None. Uncertainty: Low, Medium, High or Very High as the simulated one-year 5th-95th percentile range is under 30%, 50%, 80% of capital, or above. Projection: 400 paths of 21-day blocks drawn from the strategy's own history, with optional return shift and volatility scaling.

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Tail Risk and Quality (daily returns)

Measure	Strategy
VaR 95%, 1 day	-1.51%
CVaR 95%, 1 day	-2.22%
Normal-model VaR 95%	-1.38%
Worst day	-4.80%
Omega	1.12
Gain-to-pain	0.12
Skew	-0.59
Kurtosis (normal = 3)	6.2
Information ratio	-0.48
Tracking error	17.2%
Treynor (excess return per beta)	+21.9%
Jensen alpha (per year)	+2.0%
Profit factor	2.21

Robustness check	Result
Not run	use "Is this result real?" in the app

How to read these

VaR and CVaR: the loss on a 1-in-20 day, and the average loss on the worst 5% of days (historical, from the daily returns). Omega: total gains divided by total losses. Information ratio: active return over the benchmark divided by tracking error. Treynor: excess return per unit of beta. Probabilistic Sharpe: the chance the true Sharpe ratio is above 0, adjusting for skew and fat tails. Deflated Sharpe: the same, after charging for the number of dial settings tried. Resampling: 500 histories built from 21-day blocks of the strategy's own daily returns. First 70% / last 30%: the same fixed rules measured on each part; it checks stability over time, it does not re-tune. A passing grade lowers the odds of a lucky result and does not prove an edge.

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Stress Tests (same rules on the full history from 2002-07-31)

Scenario	Strategy	Worst dip	QQQ	Worst dip
Dot-com bust	not covered			
Global financial crisis	-16.2%	-21.0%	-51.6%	-53.4%
2011 euro-debt scare	-8.9%	-9.1%	-14.0%	-16.1%
2015-16 China slowdown	-10.9%	-12.4%	-14.8%	-16.1%
Q4 2018 selloff	-19.3%	-20.5%	-22.0%	-22.7%
COVID crash	-5.0%	-6.7%	-27.9%	-28.6%
2022 rate shock	-21.1%	-21.3%	-34.3%	-34.3%
2025 tariff shock	-10.9%	-11.3%	-22.8%	-22.8%

If QQQ fell 10% / 20% / 30%, the downside beta of 0.25 suggests about -3% / -5% / -8% for this strategy. A straight-line estimate: real crashes also move correlations.

Performance by Market Regime (read at the prior close)

Regime	Days	Strategy / yr	Sharpe	QQQ / yr	Sharpe
Uptrend (above 200-day avg)	81%	+10.2%	0.68	+14.8%	0.87
Downtrend	19%	+0.8%	0.37	+23.8%	0.69
Calm markets	33%	+6.1%	0.47	+9.6%	0.72
Normal markets	33%	+16.1%	1.07	+23.3%	1.39
Stormy markets	33%	+2.9%	0.24	+16.9%	0.55

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